

4MA1_2F_ch2_equations_formulae_identities

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1. 4MA1/2FR_Summer_2024_Q22

1 point

(a) Solve $\frac{2f}{3} = 4f - 17$

Show clear algebraic working.

$f = \dots\dots\dots$
(3)

(b) Simplify $(e + 12)^0$ where $e > 0$

(1)

(c) Simplify fully $\frac{12a^4h^6}{4ah^2}$

(2)

(d) Factorise fully $20x^5y + 12x^3y^4$

(2)

(a) Simplify $c \times c \times c$

(1)

(b) Simplify $12d \times 3e$

(1)

(c) Solve $\frac{k}{4} = 7$

$k = \dots\dots\dots$

(1)

(d) Solve $2g - 3 = 6$

$g = \dots\dots\dots$

(2)

(e) Expand $x(x - 4)$

(1)

$$P = 4y^2 + w$$

(f) Work out the value of P when $y = -3$ and $w = 2$

$P = \dots\dots\dots$

(2)

3. 4MA1/2FR_Summer_2024_Q6

1 point

- (a) Find the value of 3.4^2 (1)
- (b) Find the cube root of 373 248 (1)
- (c) Write $7 \times 7 \times 7 \times 7 \times 7 \div 7$ as a single power of 7 (1)
- (d) Write one set of brackets in each calculation to make the answer correct.
- (i) $5 + 3 \times 2 = 16$ (1)
- (ii) $10 - 8 - 10 - 6 \div 2 = 0$ (1)

4. 4MA1/2F_Summer_2024_Q27

1 point

- (i) Factorise $x^2 + 9x - 22$ (2)
- (ii) Hence, solve $x^2 + 9x - 22 = 0$ (1)

5. 4MA1/2F_Summer_2024_Q26

1 point

Solve the simultaneous equations

$$6x + 4y = 1$$

$$3x + 5y = 8$$

Show clear algebraic working.

$$x = \dots\dots\dots$$

$$y = \dots\dots\dots$$

(3)

6. 4MA1/2F_Summer_2024_Q22

1 point

Change a speed of w metres per second to a speed in kilometres per hour.

Give your answer in terms of w in its simplest form.

..... kilometres per hour

(3)

A plane takes 9 hours 36 minutes to fly from New Delhi to Perth.

The plane flies at an average speed of 820 km/h.

Work out the total distance the plane flies.

..... km

(3)

(a) Simplify $e \times e \times e \times e \times e$

(1)

(b) Simplify $m + m + m$

(1)

(c) Simplify $3g^2 + 7g^2 - 4g^2$

(1)

(d) Expand $a(a + 8)$

(1)

(e) Factorise $15x + 20$

(1)

Sophia sells d packs of toy cars and h boxes of toy cars.

Each pack contains 3 toy cars.

Each box contains 5 toy cars.

The total number of toy cars that Sophia sells is T

(f) Write down a formula for T in terms of d and h

(3)

2 jugs each contain 350 millilitres of milk.

5 bottles each contain y millilitres of milk.

Peter tips all the milk from the 2 jugs and the 5 bottles into a container.

The total amount of milk that Peter tips into the container is 2.8 litres.

Work out the value of y

(4)

(a) Expand $x(10 - x)$

.....
(1)

(b) Factorise $6y + 27$

.....
(1)

(c) Make m the subject of the formula $h = \frac{m}{2} + 4$

.....
(2)

(d) Solve $7g + 3 = 2g - 5$
Show clear algebraic working.

$g =$
(3)

(a) Write down the value of $(m + 2)^0$ where m is a positive integer.

.....
(1)

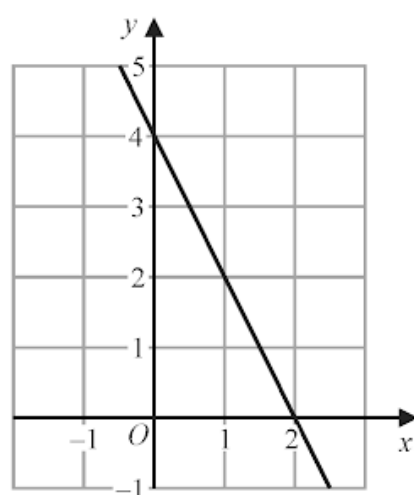
(b) Simplify $(3a^2b^4)^3$

.....
(2)

(c) Factorise fully $14x^2y^4 + 21x^3y^2$

.....
(2)

The diagram shows a straight line drawn on a grid.



(d) Write down an equation of the line.

.....
(2)

Solve the simultaneous equations

$$\begin{aligned}5x + 4y &= -2 \\ 2x - y &= 4.4\end{aligned}$$

Show clear algebraic working.

$$x = \dots\dots\dots$$

$$y = \dots\dots\dots$$

3 marks

The diagram shows rectangle $ABCD$

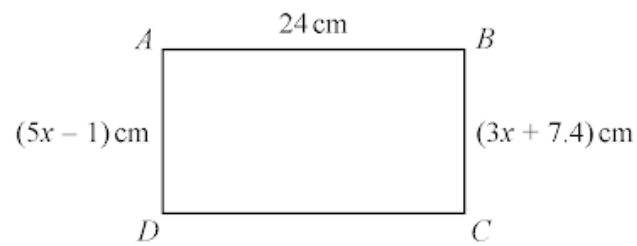


Diagram **NOT**
accurately drawn

Work out the perimeter of the rectangle.
Show your working clearly.

..... cm

4 marks

(a) Simplify $c \times c \times c \times c \times c$

.....
(1)

(b) Solve $5 + x = 12$

$x =$
(1)

(c) Solve $\frac{y}{6} = 3$

$y =$
(1)

(d) Expand $5(2 + 3h)$

.....
(1)

(e) Factorise $g^2 + 7g$

.....
(1)

(a) Simplify $(4^{-2})^0$

.....
(1)

$$3^{-14} \times 3^8 = 3^m$$

(b) Find the value of m

$m =$
(1)

(a) Factorise $6y - 27$

.....
(1)

(b) Expand $p(p - 2)$

.....
(1)

$$T = 5g + 4r$$

(c) Work out the value of r when $T = 46$ and $g = 17$

$r =$
(3)

$$P = m^2 - 4c$$

(d) Work out the value of P when $m = -5$ and $c = 3$

$P =$
(2)

(e) Expand and simplify $(x + 5)(x - 7)$

.....
(2)

(a) Expand $x(3 - x)$

.....
(1)

(b) Factorise fully $12a - 18b$

.....
(2)

There are 8 slices of cheese in each small pack of cheese.

There are 20 slices of cheese in each large pack of cheese.

Afreen buys h small packs of cheese and j large packs of cheese.

She buys a total of T slices of cheese.

(c) Write down a formula for T in terms of h and j

.....
(3)

.....

.....

.....

.....

.....

(a) Solve $4y + 5 > 12$

.....
(2)

(b) Solve $6x - 5 = \frac{4x - 7}{2}$

Show clear algebraic working.

$x =$
(3)

(a) Solve $5c = 15$

$$c = \dots\dots\dots$$

(1)

(b) Expand $x(8 - x)$

$$\dots\dots\dots$$

(1)

$$T = 5m - 6n$$

(c) Work out the value of T when $m = 4.2$ and $n = -2.5$

$$T = \dots\dots\dots$$

(2)

(d) Make g the subject of $k = 2g + t$

$$\dots\dots\dots$$

(2)

Solve the simultaneous equations

$$3x + 5y = 3.1$$

$$6x + 3y = 3.75$$

Show clear algebraic working.

$$x = \dots\dots\dots$$

$$y = \dots\dots\dots$$

3 marks

(a) Expand and simplify $x(2x - 3) + 7(2x + 1) - 5$

.....
(3)

(b) Expand and simplify $(y + 4)(2 - y)$

.....
(2)

(c) Factorise fully $15b^5c - 35b^3c^9$

.....
(2)

$$\frac{2^k}{4^n} = 2^x$$

Find an expression for x in terms of k and n

$$x = \dots\dots\dots$$

2 marks

(a) Solve $p = \frac{3p-5}{10}$

Show clear algebraic working.

$$p = \dots\dots\dots$$

(3)

(b) Simplify a^0 where $a > 0$

$$\dots\dots\dots$$

(1)

(c) Simplify fully $\frac{3xy^3}{6x^2y}$

$$\dots\dots\dots$$

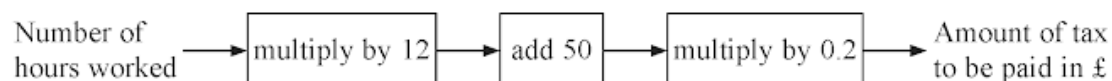
(2)

(d) Factorise fully $10c^3d^2 + 15cd^4$

$$\dots\dots\dots$$

(2)

Molly uses this number machine to work out the amount of tax that she has to pay on the money she earns.



When Molly works n hours the amount of tax she has to pay is £ T

Find a formula for T in terms of n

.....

3 marks

(a) Simplify $a \times a \times a \times a \times a$

.....
(1)

(b) Simplify $8b \times 3c$

.....
(1)

(c) Expand $3(x + 4)$

.....
(1)

$$Q = 5v^2 - w$$

(d) Work out the value of Q when $v = \frac{1}{2}$ and $w = \frac{1}{4}$

$Q =$
(2)

(i) Factorise $x^2 + 5x - 24$

.....
(2)

(ii) Hence, solve $x^2 + 5x - 24 = 0$

.....
(1)

Solve the simultaneous equations

$$7x + 3y = 3$$

$$3x - y = 7$$

Show clear algebraic working.

$$x = \dots\dots\dots$$

$$y = \dots\dots\dots$$

(a) Simplify $c \times c \times c \times c \times c \times c$

.....
(1)

(b) Simplify $2h^3 + 5h^3 - h^3$

.....
(1)

(c) Expand $x(x + 5)$

.....
(1)

(d) Factorise $9y - 12$

.....
(1)

Rosanna sells m small bags of marbles and p large bags of marbles.

Each small bag contains 15 marbles.

Each large bag contains 40 marbles.

The total number of marbles that Rosanna sells is T

(e) Write down a formula for T in terms of m and p

.....
(3)

.....

.....

.....

.....

.....

Solve $x^2 - 21x + 20 = 0$
Show your working clearly.

3 marks

(a) Simplify $(3k^2)^4$

.....
(2)

(b) Simplify $(21m^4n) \div (3n^{-5})$

.....
(2)

Solve $5(2x - 3) = 20$
Show clear algebraic working.

$x =$

3 marks

$$w = 5y^2 - y^3$$

(a) Work out the value of w when $y = -2$

$$w = \text{.....}$$

(2)

(b) Factorise fully $8p^2 - 2p$

$$\text{.....}$$

(2)

(c) Expand $4t(3t - 2)$

$$\text{.....}$$

(2)

(d) Expand and simplify $(5x - 2)(x + 4)$

$$\text{.....}$$

(2)

(a) Expand and simplify $4x(2x + 5) - 3x(2x - 3)$

.....
(2)

Given that $\frac{y^5 \times y^n}{y^6} = y^{13}$

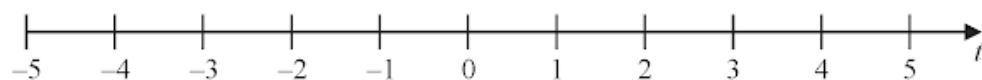
(b) work out the value of n .

$n =$
(2)

(c) (i) Solve the inequality $7t - 8 < 2t + 7$

.....
(2)

(ii) On the number line below, represent the solution set of the inequality solved in part (c)(i)



(1)

The diagram shows a right-angled triangle.

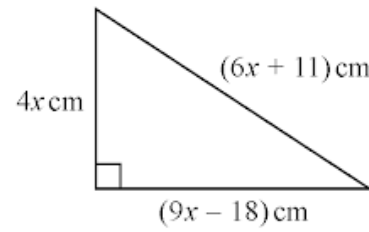


Diagram **NOT**
accurately drawn

The perimeter of the triangle is 126 cm.

Work out the area of the triangle.

..... cm^2

4 marks

(a) Simplify $p + p + p + p + p + p$

.....
(1)

(b) Simplify $5y^2 + 6y^2 - 3y^2$

.....
(1)

(c) Simplify $e \times e \times e \times e \times e$

.....
(1)

(d) Simplify $5c \times 4d$

.....
(1)

(e) Solve $x - 7 = 19$

$x =$
(1)

$$18^2 + 15^2 - 5^3 = 4n$$

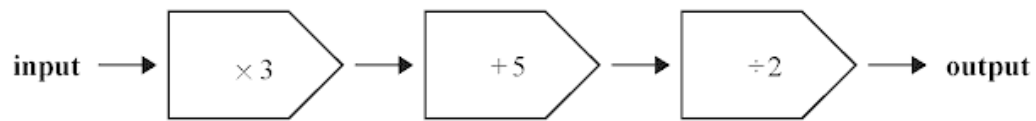
(f) Work out the value of n .

$n =$
(2)

(g) Factorise $9t - 6$

.....
(1)

Here is a number machine.



(a) Work out the output when the input is 7

.....
(1)

(b) Work out the input when the output is 160

.....
(2)

When the input is n , the output is P .

(c) Find a formula for P in terms of n .

.....
(2)

.....

.....

.....

.....

.....

(i) Factorise $x^2 + 2x - 24$

.....
(2)

(ii) Hence solve $x^2 + 2x - 24 = 0$

.....
(1)

Solve the simultaneous equations

$$5a + 2c = 10$$

$$2a - 4c = 7$$

Show clear algebraic working.

$$a = \dots\dots\dots$$

$$c = \dots\dots\dots$$

3 marks

(a) Simplify $w \times w \times w \times w \times w$

.....
(1)

(b) Simplify $5a \times 3c$

.....
(1)

(c) Simplify $3e + 2f - e + 5f$

.....
(2)

(d) Solve $5x - 7 = x + 12$
Show clear algebraic working.

$x =$
(3)

The diagram shows a box in the shape of a cuboid.

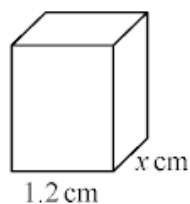


Diagram **NOT**
accurately drawn

The box is put on a table.

The face of the box in contact with the table has length 1.2 metres and width x metres.

The force exerted by the box on the table is 27 newtons.

The pressure on the table due to the box is 30 newtons/m²

$$\text{pressure} = \frac{\text{force}}{\text{area}}$$

Work out the value of x .

$x =$

3 marks

Solve the simultaneous equations

$$3x + 5y = 6$$

$$7x - 5y = -11$$

Show clear algebraic working.

$$x = \dots\dots\dots$$

$$y = \dots\dots\dots$$

3 marks

(a) Expand $e(3e - 5)$

.....
(1)

(b) Factorise $35 + 5f$

.....
(1)

(c) Simplify $(4pq^2)^3$

.....
(2)

(a) Simplify $k + k + k + k$

$$f = 9 \times 9 \times 9 \times 9$$

(b) (i) Write f as a single power of 9

.....
(1)

(ii) Write f as a single power of 3

.....
(2)

(c) Write $5^{17} \times 5^2$ as a single power of 5

.....
(1)

(d) Write 800 as a product of its prime factors.
Show your working clearly.

.....
(2)

.....

.....

.....

.....

.....

(a) Solve the inequality $2x + 7 > 4$

(2)

(b) Solve $x^2 - 3x - 40 = 0$
Show clear algebraic working.

(3)

The diagram shows a triangle.

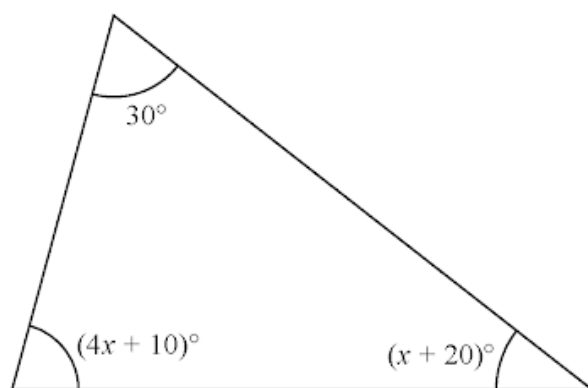


Diagram **NOT**
accurately drawn

Work out the value of x .

$x =$

4 marks

(a) Simplify $\frac{x^9}{x^2}$

(1)

(b) Write $\frac{7^8 \times 7^4}{7^3}$ as a single power of 7

(2)

(a) Expand $x(5 - x)$

.....
(1)

(b) Factorise $3y - 21$

.....
(1)

(c) Make p the subject of the formula $f = 3p - d$

.....
(2)

Sergio buys m boxes of seeds and n packets of seeds.

Each box contains 10 seeds.

Each packet contains 6 seeds.

The total number of seeds that Sergio buys is T .

(d) Write down a formula for T in terms of m and n .

.....
(3)

.....

.....

.....

.....

.....

(a) Simplify $10a \times b$

.....
(1)

(b) Solve $n + 3 = 7$

$n =$
(1)

Find the gradient of the straight line with equation $5x + 2y = 7$

.....
2 marks

Given that $150^x = 1$

(a) write down the value of x .

$x = \dots\dots\dots$
(1)

Given that $3^{-8} \div 3^{-6} = 3^n$

(b) find the value of n .

$n = \dots\dots\dots$
(1)

53. 4MA1/2F_Summer_2020_Q21 1 point

54. 4MA1/2F_Summer_2020_Q14 1 point

55. 4MA1/2FR_Winter_2019_Q25 1 point

56. 4MA1/2FR_Winter_2019_Q13 1 point

57. 4MA1/2FR_Winter_2019_Q11

1 point

58. 4MA1/2FR_Winter_2019_Q7

1 point

59. 4MA1/2F_Winter_2019_Q23

1 point

60. 4MA1/2F_Winter_2019_Q11

1 point

61. 4MA1/2F_Winter_2019_Q4 1 point

62. 4MA1/2FR_Summer_2019_Q17 1 point

63. 4MA1/2FR_Summer_2019_Q13 1 point

64. 4MA1/2F_Summer_2019_Q20 1 point

65. 4MA1/2F_Summer_2019_Q9 1 point

66. 4MA1/2FR_Summer_2018_Q21 1 point

67. 4MA1/2FR_Summer_2018_Q19 1 point

68. 4MA1/2FR_Summer_2018_Q15 1 point

69. 4MA1/2F_Summer_2018_Q24 1 point

70. 4MA1/2F_Summer_2018_Q23 1 point

71. 4MA1/2F_Summer_2018_Q18 1 point

72. 4MA1/2F_Summer_2018_Q8 1 point

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